

Shardul Kattewar

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SUMMARY

Detail-oriented Backend Engineer and CSBS student specializing in high-performance Python application architectures, data pipelines, and cryptographic security. Proven track record of developing autonomous financial agents and secure algorithmic trading engines with an emphasis on modular, fail-safe system design

EDUCATION

D.Y Patil , RAIT, 2023 – 2027 | Nerul
B.Tech in Computer Science & Business Systems

- **Current CGPA: 7.32/10** (Consistent improvement from 4.0 in Semester 1; **83% growth**)
- **Relevant Coursework:** Data Structures & Algorithms (DSA), Relational Database Management Systems (RDBMS), Applied Statistics, Information Security.

TECHNICAL SKILLS

Languages: — Python (Data Science focus), C++ (DSA & Logic), C, SQL (SQLite), JavaScript,
Libraries/Frameworks: — NumPy, Pandas, Scikit-learn, Matplotlib, **Developer Tools** — **Backend & Web:** REST APIs, HMAC-SHA256 Cryptography, Node.js / React (for FinReAct), Ollama, Llama 3, Agentic ReAct Architecture, Prompt Engineering

CERTIFICATES

- Advanced Data Science & Machine Learning | iHUB IIT Roorkee — Predictive modeling, data pipelines, and architectural frameworks.
- Microsoft SQL Technical Certification | Microsoft — Advanced database management, relational indexing, and query optimization.

PROJECTS

Binance Futures Algorithmic Trading Engine | Python, REST API,

- Engineered a modular CLI trading bot in Python to securely execute live MARKET/LIMIT orders on the Binance Testnet using signed SHA256 HMAC encryption.
- Designed a fail-safe input boundary validation framework to intercept payload anomalies and network faults, minimizing API runtime errors.
- Developed an interactive multi-step terminal UI wizard with live telemetry tracking to streamline manual contract parameter configurations.

FinReact – Autonomous Local AI Financial Agent | Python, SQL, SQLite, Ollama, Llama 3

- **Hybrid ReAct Agent Architecture:** Engineered a local-first autonomous financial agent using **Python** and an offline **Llama 3** model, ensuring 100% data privacy for sensitive personal portfolio transactions.
- **Intent Routing Layer:** Implemented a custom routing mechanism that dynamically evaluates user prompts to intelligently branch execution between local relational database context and autonomous web scraping tools.
- **Text-to-SQL Engine:** Designed an object-oriented translation pipeline that dynamically parses natural language questions into valid, optimized **SQL** queries to extract data from a structured SQLite binary database.
- **Stateful Memory Management:** Optimized multi-turn conversation context by implementing a **Sliding-Window Memory algorithm**, successfully capping prompt payloads to maintain low local processing latency.

AI-Driven Precision Agriculture & Crop Recommendation, *Developed a predictive model to optimize agricultural output based on soil chemistry and environmental factors.*

- Implemented **Random Forest** and **XGBoost** models using Python and Scikit-learn.
- Engineered a data pipeline to process NPK (Nitrogen, Phosphorus, Potassium) and climate data.
- Refactored prompt-generated code into **Modular OOP structures** to ensure scalability and code maintainability.
- Used **Cursor** for rapid prototyping and local environment management.

Agentic AI Portfolio & Research Hub, *React, Tailwind CSS, Claude 3.5 Sonnet*

- Engineered a professional academic portfolio using an **Agentic AI workflow**, focusing on modular component architecture.
- Implemented a custom **"AI Transparency"** disclosure to document the human-AI collaborative development process.
- Integrated live links to research dossiers, academic marksheets, and predictive ML project reports.